

Week 3 In-Class Worksheet - ANSWER KEY

Probability Distributions and Sampling Distributions

Course: H524 Introduction to Biostatistics

Topic: Binomial, Normal, Sampling Distributions, and Confidence Intervals

1 Problem 1: Binomial Distribution

A new antibiotic has an 80% success rate in treating a particular bacterial infection. A clinic treats 15 patients with this antibiotic.

Part A: What is the probability that exactly 12 patients are successfully treated?

Given: $n = 15$, $p = 0.80$, find $\Pr(X = 12)$

Formula: $\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$

Solution:

$$\begin{aligned}\Pr(X = 12) &= \binom{15}{12} (0.80)^{12} (0.20)^3 \\ &= \frac{15!}{12!3!} (0.80)^{12} (0.20)^3 \\ &= 455 \times 0.06872 \times 0.008 \\ &= \mathbf{0.2501}\end{aligned}$$

Part B: What is the probability that at least 13 patients are successfully treated?

Hint: $\Pr(X \geq 13) = \Pr(X = 13) + \Pr(X = 14) + \Pr(X = 15)$

Solution:

$$\begin{aligned}\Pr(X = 13) &= \binom{15}{13} (0.80)^{13} (0.20)^2 = 0.2309 \\ \Pr(X = 14) &= \binom{15}{14} (0.80)^{14} (0.20)^1 = 0.1319 \\ \Pr(X = 15) &= \binom{15}{15} (0.80)^{15} (0.20)^0 = 0.0352 \\ \Pr(X \geq 13) &= 0.2309 + 0.1319 + 0.0352 = \mathbf{0.3980}\end{aligned}$$

Part C: Calculate the expected number of successful treatments and the standard deviation.

Expected value: $\mu = np$

Solution: $\mu = 15 \times 0.80 = \mathbf{12}$

Standard deviation: $\sigma = \sqrt{np(1-p)}$

Solution: $\sigma = \sqrt{15 \times 0.80 \times 0.20} = \sqrt{2.4} = \mathbf{1.549}$

Part D: Use R to verify your answers. Write the R code you would use:

Solution:

```
# Part A:  $P(X = 12)$ 
dbinom(12, size=15, prob=0.80)

# Part B:  $P(X \geq 13)$ 
sum(dbinom(13:15, 15, 0.80))
# OR: pbinom(12, 15, 0.80, lower.tail=FALSE)

# Part C: Expected value and SD
n <- 15
p <- 0.80
expected <- n * p
std_dev <- sqrt(n * p * (1-p))
```

2 Problem 2: Normal Distribution

Adult systolic blood pressure is normally distributed with mean 125 mmHg and standard deviation 18 mmHg.

Part A: What proportion of adults have blood pressure above 140 mmHg (hypertension)?

Step 1: Standardize using $Z = \frac{X - \mu}{\sigma}$

Solution:

$$Z = \frac{140 - 125}{18} = \frac{15}{18} = 0.833$$

Step 2: Find $\Pr(Z > z)$ using the standard normal table or R

Solution:

$$\Pr(Z > 0.833) = 1 - \Pr(Z < 0.833) = 1 - 0.7976 = \mathbf{0.2024}$$

About **20.2%** of adults have blood pressure above 140 mmHg.

Part B: What proportion have blood pressure between 110 and 140 mmHg?

Hint: $\Pr(110 < X < 140) = \Pr(Z < z_2) - \Pr(Z < z_1)$

Solution:

$$Z_1 = \frac{110 - 125}{18} = -0.833$$

$$Z_2 = \frac{140 - 125}{18} = 0.833$$

$$\begin{aligned}\Pr(110 < X < 140) &= \Pr(Z < 0.833) - \Pr(Z < -0.833) \\ &= 0.7976 - 0.2024 = \mathbf{0.5952}\end{aligned}$$

About **59.5%** have blood pressure in this range.

Part C: Find the blood pressure value that represents the 90th percentile.

Step 1: Find z such that $\Pr(Z < z) = 0.90$

Solution: From Z-table or R: $z_{0.90} = 1.282$

Step 2: Convert back to original scale: $X = \mu + z\sigma$

Solution:

$$X = 125 + 1.282(18) = 125 + 23.08 = \mathbf{148.08} \text{ mmHg}$$

Part D: Write the R code to solve parts A, B, and C:

Solution:

```
mu <- 125
sigma <- 18

# Part A: P(X > 140)
pnorm(140, mu, sigma, lower.tail=FALSE)

# Part B: P(110 < X < 140)
pnorm(140, mu, sigma) - pnorm(110, mu, sigma)

# Part C: 90th percentile
qnorm(0.90, mu, sigma)
```

3 Problem 3: Sampling Distribution and Central Limit Theorem

A population of cholesterol levels has mean $\mu = 200$ mg/dL and standard deviation $\sigma = 40$ mg/dL. A sample of $n = 64$ individuals is randomly selected.

Part A: What is the distribution of the sample mean $\bar{X}$?

By the Central Limit Theorem: $\bar{X} \sim N(\mu_{\bar{X}}, \sigma_{\bar{X}}^2)$

Mean of $\bar{X}$: $\mu_{\bar{X}} =$

Solution: $\mu_{\bar{X}} = \mu = 200$ mg/dL

Standard error of $\bar{X}$: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} =$

Solution: $\sigma_{\bar{X}} = \frac{40}{\sqrt{64}} = \frac{40}{8} = 5$ mg/dL

Therefore: $\bar{X} \sim N(200, 5^2)$ or $\bar{X} \sim N(200, 25)$

Part B: What is the probability that the sample mean is greater than 210 mg/dL?

Step 1: Standardize: $Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$

Solution:

$$Z = \frac{210 - 200}{5} = 2.0$$

Step 2: Find probability

Solution:

$$\Pr(\bar{X} > 210) = \Pr(Z > 2.0) = 1 - 0.9772 = \mathbf{0.0228}$$

R code: `pnorm(210, 200, 5, lower.tail=FALSE)`

Part C: What is the probability that the sample mean is between 195 and 205 mg/dL?

Solution:

$$Z_1 = \frac{195 - 200}{5} = -1.0$$

$$Z_2 = \frac{205 - 200}{5} = 1.0$$

$$\begin{aligned}\Pr(195 < \bar{X} < 205) &= \Pr(-1.0 < Z < 1.0) \\ &= \Pr(Z < 1.0) - \Pr(Z < -1.0) \\ &= 0.8413 - 0.1587 = \mathbf{0.6826}\end{aligned}$$

R code: `pnorm(205, 200, 5) - pnorm(195, 200, 5)`

Part D: How would your answers change if $n = 16$ instead of $n = 64$? Would that probability increase or decrease? Explain why.

Solution:

With $n = 16$: $\sigma_{\bar{X}} = \frac{40}{\sqrt{16}} = 10$ (larger SE)

That probability would **increase**. With a smaller sample size:

- Standard error is larger (10 vs. 5)
- The sampling distribution is more spread out
- Extreme values (like 210) are more likely
- Z-score would be smaller: $Z = \frac{210-200}{10} = 1.0$ instead of 2.0
- $\Pr(Z > 1.0) = 0.1587$ vs. $\Pr(Z > 2.0) = 0.0228$

4 Problem 4: Confidence Intervals

A clinical trial measures the reduction in pain scores for 20 patients receiving a new medication. The sample mean reduction is $\bar{x} = 4.5$ points, with sample standard deviation $s = 1.8$ points.

Part A: Calculate a 95% confidence interval for the true mean pain reduction.

Step 1: Determine what critical value to use. Should you use z or t ? Why?

Solution:

Use **t-distribution** because:

- Population SD (σ) is unknown
- Small sample size ($n = 20 < 30$)

Step 2: Find the critical value.

Degrees of freedom = $n - 1 =$

Solution: $df = 20 - 1 = 19$

Critical value: $t_{0.975, df} =$ (use R: `qt(0.975, df)`)

Solution: $t_{0.975, 19} = \mathbf{2.093}$

Step 3: Calculate the standard error: $SE = \frac{s}{\sqrt{n}}$

Solution:

$$SE = \frac{1.8}{\sqrt{20}} = \frac{1.8}{4.472} = \mathbf{0.403}$$

Step 4: Calculate the margin of error: $ME = t_{critical} \times SE$

Solution:

$$ME = 2.093 \times 0.403 = \mathbf{0.843}$$

Step 5: Construct the confidence interval: $\bar{x} \pm ME$

Solution:

$$CI = 4.5 \pm 0.843 = (\mathbf{3.66}, \mathbf{5.34}) \text{ points}$$

Part B: Interpret the confidence interval in context. Write a complete sentence explaining what this interval means.

Solution:

We are 95% confident that the true mean pain reduction for all patients receiving this medication is between 3.66 and 5.34 points.

Part C: If you wanted a 99% confidence interval instead, would it be wider or narrower than the 95% interval? Explain.

Solution:

The 99% CI would be **wider**. To be more confident that we've captured the true mean, we need a wider interval. The critical value increases ($t_{0.995, 19} = 2.861$ vs. $t_{0.975, 19} = 2.093$), resulting in a larger margin of error.

5 Problem 5: Integration Problem

A pharmaceutical company is testing a new cholesterol-lowering drug. In the clinical trial:

- 100 patients receive the drug
- Cholesterol reduction follows a normal distribution
- Sample mean reduction: $\bar{x} = 28$ mg/dL
- Sample standard deviation: $s = 12$ mg/dL

Part A: Calculate a 95% confidence interval for the true mean cholesterol reduction.

Solution:

With large sample ($n = 100$), use z-distribution: $z_{0.975} = 1.96$ (or t-distribution: $t_{0.975,99} \approx 1.984$)

$$SE = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{100}} = 1.2$$

$$ME = 1.96 \times 1.2 = 2.352$$

$$CI = 28 \pm 2.352 = (25.65, 30.35) \text{ mg/dL}$$

Part B: Based on your confidence interval, is there strong evidence that the drug reduces cholesterol by more than 25 mg/dL on average? Explain your reasoning.

Solution:

Yes, there is strong evidence. The entire 95% confidence interval (25.65, 30.35) is above 25 mg/dL. This means we can be 95% confident that the true mean reduction exceeds 25 mg/dL.

Part C: If individual cholesterol reductions are normally distributed with the sample statistics above, approximately what proportion of patients will experience a reduction of at least 40 mg/dL?

Hint: Use the sample statistics as estimates of population parameters.

Solution:

Using $\mu = 28$ and $\sigma = 12$ as population estimates:

$$Z = \frac{40 - 28}{12} = 1.0$$

$$\Pr(X \geq 40) = \Pr(Z \geq 1.0) = 1 - 0.8413 = \mathbf{0.1587}$$

About **15.9%** of patients will experience a reduction of at least 40 mg/dL.

R code: `pnorm(40, 28, 12, lower.tail=FALSE)`

Part D: Explain the difference between the questions in Parts B and C. Why do we use different approaches (confidence interval vs. normal probability)?

Solution:

Part B (Confidence Interval): Asks about the *population mean* - we use the sampling distribution of the sample mean, which has $SE = \sigma/\sqrt{n} = 1.2$.

Part C (Normal Probability): Asks about *individual patients* - we use the population distribution with $SD = \sigma = 12$.

The key difference:

- Part B: Inference about a parameter (mean) $\rightarrow$ use SE
- Part C: Prediction about individuals $\rightarrow$ use SD

Sample means are much less variable than individual observations, so the CI is narrower than the range containing 95% of individuals.

Reflection Questions

1. What is the key difference between a standard deviation and a standard error?

Answer:

Standard Deviation (SD): Measures variability of individual observations around the mean. Describes the spread of the data.

Standard Error (SE): Measures variability of sample means around the population mean. Describes uncertainty in our estimate of the mean. $SE = \sigma/\sqrt{n}$, so it decreases with larger sample sizes.

2. When should you use a t -distribution instead of a normal distribution for confidence intervals?

Answer:

Use t -distribution when:

- Population standard deviation (σ) is unknown (must use sample SD s)
- Sample size is small (typically $n < 30$)

With large samples ($n \geq 30$), the t -distribution closely approximates the normal distribution.

3. How does sample size affect the width of a confidence interval?

Answer:

As sample size increases, confidence intervals become **narrower** (more precise):

- Larger $n \rightarrow$ smaller SE ($SE = s/\sqrt{n}$)
- Smaller SE $\rightarrow$ smaller margin of error
- Smaller ME $\rightarrow$ narrower CI

To cut the CI width in half, you need to quadruple the sample size.